Temporal Genesis: A Self-Contained Master Document

A simple, intuitive, and mathematically consistent story

Compiled Synthesis

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Abstract

We synthesize a complete, self-contained narrative—*temporal genesis*—that turns a timeless description into operational dynamics using signal processing, matrix identities, and covariant limits. The spine is: (i) an exact two-step recursion on decimated residue tracks (Cayley–Hamilton), (ii) a compact *wheel* dictionary that equates this recursion to a dominant conjugate pole pair, (iii) a *Mirror–Outer Spectral Content* (MOSC) operator that preserves spectral magnitude while stripping all-pass phase, (iv) an operational dephasing index tied to phase-increment noise, (v) a binary *APBC* flip mechanism for measurement and Born statistics, and (vi) an emergent Lorentz sector with a dynamic envelope whose strength depends on the decoherence field, producing an effective metric. All non-standard terms are defined; key claims are proved or sketched. We include diagnostics, a practitioner pipeline, and appendices with derivations and ray-tracing pseudo-code.

1 Roadmap and executive summary

From conditioning to a two-step law. Condition a timeless global state on a clock sector. On decimated residue tracks one obtains an *exact* recursion

$$y_{k+2} - V_\ell\, y_{k+1} + Q_\ell\, y_k = 0, \quad (V_\ell, Q_\ell) = (\mathrm{tr} M^\ell,\ (\det M)^\ell), \tag{1}$$

where $M \in \mathbb{C}^{2\times 2}$ is a reduced transfer and $\ell \in \mathbb{Z}_{>0}$ a stride.

Wheel (dominant pair). If $r_\pm = \rho e^{\pm i\theta}$ dominates, then

$$(V, Q) = (2\rho\cos\theta,\ \rho^2), \qquad V_\ell = 2\rho^\ell \cos(\ell\theta), \quad Q_\ell = \rho^{2\ell}. \tag{2}$$

MOSC. Factor $H = O \cdot I$ (outer $\times$ inner/all-pass) and define $\mathcal{S}[H] := H/I$. Then $|\mathcal{S}[H]| = |H|$ on the unit circle; $\mathcal{S}[H]$ is outer (minimum-phase).

Coherence & dephasing. Demodulate by r_+, form wrapped phase increments δ_t. Coherence decays as

$$|\mathcal{C}(t)| \approx \rho^t\, e^{-Dt}, \quad D = \tfrac{1}{2} S_\delta(0). \tag{3}$$

Measurement as APBC flips. Micro-absorbers flip PBC→APBC with hazard $\lambda \propto |\phi|^2$, marking $V \mapsto -V$ and yielding Born statistics via first-arrival. Which-path visibility $\sim e^{-D_{\text{proper}} T_{\text{proper}}}$.

Lorentz sector and dynamic envelope. The coherent AR(2) sector has a KG limit. A dynamic envelope $s = s_0 + \alpha S_\delta$ renormalizes the local wave speed, defining an effective metric g and closing a self-consistent gravity–locality loop.

Gravity map. $\Phi \propto S_\delta(0)$ gives quasi-static screened Poisson; dynamics follow a telegraph/KG closure with speed $\leq v_{\max}$ and screening length tied to s.

2 Preliminaries, notation, and standard background

Global conventions and harmonization

Metric/sign. We use the "mostly minus" convention $\eta = \mathrm{diag}(1, -1, -1, -1)$ and the d'Alembertian $\Box := \frac{1}{c^2}\partial_t^2 - \nabla^2$.

Mass/frequency. The Klein–Gordon mass parameter is m; earlier frequency notation relates via $m \equiv \omega_0/c$.

Proper-time dephasing. Define $\mathcal{S}(x)$ per proper time; $D_{\text{proper}} = \frac{1}{2}\mathcal{S}$. Under a constant boost with dilation factor γ, the coordinate-time dephasing rate scales as $D = \gamma D_{\text{proper}}$ (the zero-frequency PSD scales covariantly under time stretching), so $|\mathcal{C}(t)| \sim e^{-Dt}$ becomes $e^{-\gamma D_{\text{proper}} t}$.

Effective metric. A dynamic envelope yields $c_{\text{eff}}^2 = c^2 - 2\sigma$ with $\sigma = s/\tau^2$, producing an effective metric $g^{00} = 1$, $g^{ij} = -(c_{\text{eff}}^2/c^2)\delta^{ij}$ in which the principal symbol is Lorentzian.

2.1 Signals, z-transform, and LTI systems

We consider complex, discrete-time signals $x : \mathbb{Z} \to \mathbb{C}$, with z-transform $X(z) = \sum_{t\in\mathbb{Z}} x_t z^{-t}$. A causal, stable LTI system has transfer $H(z)$ analytic on $|z| \geq 1$. For AR(p),

$$H(z) = \frac{\sigma}{A(z)} = \frac{\sigma}{1 - a_1 z^{-1} - \cdots - a_p z^{-p}}, \tag{4}$$

with poles p_j inside the unit disk for stability.

2.2 Hardy spaces, inner/outer, and all-pass

Let $\mathcal{H}^2$ be analytic functions on the unit disk with square-integrable boundary values; $\mathcal{H}^\infty$ are bounded functions. An *inner* $I \in \mathcal{H}^\infty$ has $|I(e^{i\omega})| = 1$ a.e.; an *outer* $O \in \mathcal{H}^2$ has logarithmic magnitude averaging property. Every nonzero $H \in \mathcal{H}^2$ factors uniquely (up to a unimodular constant) as $H = O \cdot I$.

2.3 Blaschke products (all-pass building blocks)

For $|p| < 1$, $B_p(z) = \frac{1-\bar{p}z^{-1}}{1-pz^{-1}}$ is inner; finite products build the inner factor.

2.4 Cayley–Hamilton for 2×2 matrices

Every $M \in \mathbb{C}^{2\times 2}$ satisfies $M^2 - (\mathrm{tr}M)M + (\det M)\mathrm{Id} = 0$. The same holds for M^ℓ.

2.5 Page–Wootters conditioning and residue tracks

We adopt a Page–Wootters view in which a timeless state is conditioned on a clock sector, yielding an ordered stream. For stride ℓ, residue tracks are decimated sequences $y_k^{(r)} = x_{r+k\ell}$. A *reduced transfer* M acts on a two-dimensional sufficient state across decimated steps.

3 Exact two-step law from Cayley–Hamilton

Applying Cayley–Hamilton to M^ℓ yields the exact recursion (1) for any linear readout on residue tracks.

4 The wheel: AR(2) dictionary for a dominant conjugate pair

If a conjugate pair $r_\pm = \rho e^{\pm i\theta}$ dominates, then (2) holds. The stride-ℓ versions follow as shown.

5 Mirror–Outer Spectral Content (MOSC)

Definition 5.1 (MOSC)**.** For stable rational H with inner I (finite Blaschke product) and outer O, define

$$\mathcal{S}[H] \coloneqq \frac{H}{I}. \tag{5}$$

Proposition 5.2 (Magnitude preservation)**.** $|\mathcal{S}[H](e^{i\omega})| = |H(e^{i\omega})|$ *a.e.*

Proposition 5.3 (Outer representative)**.** $\mathcal{S}[H]$ *is outer; for* $H = O \cdot I$, $\mathcal{S}[H] = O$ *up to a unit modulus constant.*

5.1 Abel/radial regularization and Ω-compatibility

Let $c_k = \mathrm{Cov}(u_t, u_{t+k})$. The Abel-regularized zero-frequency spectrum is $S_u(0) = \lim_{r\uparrow 1} \sum_k c_k r^{|k|}$. Since $|\mathcal{S}[H]| = |H|$ on the unit circle, Abel-regularized spectral magnitudes coincide.

6 Coherence & decoherence: AR(2) core of AR(∞)

Let γ_k be MOSC-filtered autocovariances. The order-2 Yule–Walker equations give the best AR(2) predictor coefficients

$$V = \frac{\gamma_1(\gamma_0 - \gamma_2)}{\gamma_0^2 - \gamma_1^2}, \qquad Q = \frac{\gamma_1^2 - \gamma_0\gamma_2}{\gamma_0^2 - \gamma_1^2}. \tag{6}$$

Coherence metrics: explained fraction $\mathcal{C}_2 = 1 - \epsilon_2^2/\gamma_0$; spectral effective order $N_{\text{eff}} = 1/\sum_k P_k^2$; dephasing $D = \frac{1}{2}S_\delta(0)$, $T_{\text{coh}} = \tau/(-\ln\rho + D)$.

7 Superposition in AR(2) and temporal lensing in AR(p)

7.1 Why AR(2) exhibits superposition

Proposition 7.1. *For $y_{t+1} - Vy_t + Qy_{t-1} = 0$ with distinct roots $r_\pm$, $y_t = c_+ r_+^t + c_- r_-^t$. For real data and $r_\pm = \rho e^{\pm i\theta}$, $y_t = A\,\rho^t\cos(\theta t + \varphi)$.*

7.2 Spatial coupling and Green-function superposition

Promoting V to $V(\Delta_a) = 2 + \tau^2(c^2\Delta_a - m^2)$ with $Q = 1$ gives the discrete KG update whose continuum limit is KG; solutions superpose through the retarded Green function.

7.3 AR(p) residues and temporal lensing

For stable AR(p), $h_t = \sum_j b_j p_j^t$. Outer moduli $|p_j| < 1$ damp; the inner factor creates group-delay corrugation that can focus energy at selected ticks (temporal lensing).

8 Operational dephasing and coherence decay

Demodulate by r_+, write $z_t = A_t e^{i\phi_t}$, and define $\delta_t = \text{wrap}(\phi_{t+1} - \phi_t)$. Under weak dependence,

Theorem 8.1. *If $S_\delta(0)$ exists (Abel-regularized), then $|\mathcal{C}(t)| \approx \rho^t e^{-Dt}$ with $D = \frac{1}{2}S_\delta(0)$.*

9 Measurement as APBC flips & the double-slit

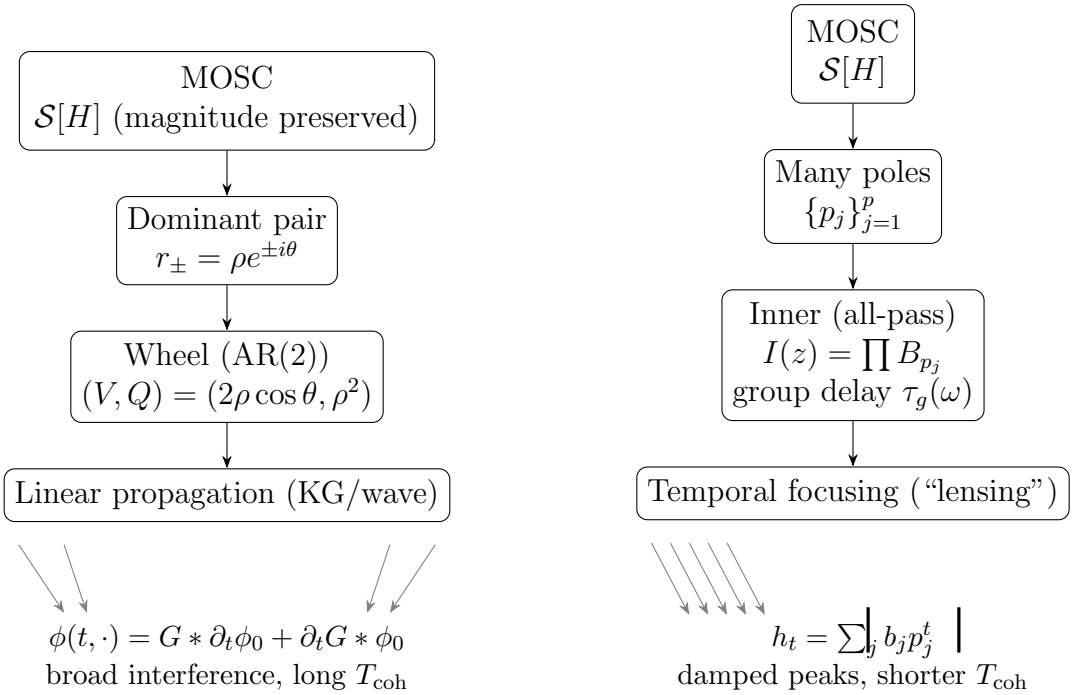
Micro-absorbers indexed by i at x_i carry two states: PBC/APBC with $\sigma_i = \pm 1$. A PBC$\to$APBC flip imposes a π mark $V \mapsto -V$, Q unchanged; flips inject poles and increase local S_δ (*topological decoherence*). For small flux, use the hazard

$$\lambda_i^+(t) = \kappa_i|\phi(t, x_i)|^2, \qquad \lambda_i^- \approx 0. \tag{7}$$

First-arrival yields Born statistics $p(x) \propto \kappa(x)\int|\phi(t,x)|^2 dt$. For a double slit with $\phi = \phi_1 + \phi_2$,

$$p(x) \;\propto\; \kappa(x)\Big(|\phi_1|^2 + |\phi_2|^2 + 2|\phi_1\phi_2|\cos\Delta\varphi\Big). \tag{8}$$

Which-path tagging suppresses the cross term by $\chi \approx e^{-D_{\text{proper}}T_{\text{proper}}}$; in a stationary detector $D = D_{\text{proper}}$, $T = T_{\text{proper}}$.



(a) AR(2) coherent superposition

(b) AR(p) temporal lensing

Figure 1: AR(2) coherent superposition vs. AR(p) temporal lensing.

10 Lorentz invariance: status, conditions, and proof-of-principle

10.1 Operational definition and status

We take $\eta = \mathrm{diag}(1, -1, -1, -1)$ and $\Box = \frac{1}{c^2}\partial_t^2 - \nabla^2$. The full discrete pipeline is not exactly LI (spatial envelope, telegraph term, coordinate-time dephasing), but the coherent AR(2) core has an LI continuum limit.

10.2 Sufficient conditions and theorem

Assume unitary wheel (Q=1), isotropic local coupling with dispersion $\omega^2(\boldsymbol{k}) = c^2\|\boldsymbol{k}\|^2 + m^2 + \mathcal{O}(a^2\|\boldsymbol{k}\|^4)$, and a vanishing/covariant regulator. Then:

Theorem 10.1 (AR(2)$\Rightarrow$ KG)**.** *As $a, \tau \to 0$ with $a/\tau \to c$, the operator-valued AR(2) update $\phi_{t+\tau} - V(\Delta_a)\phi_t + \phi_{t-\tau} = 0$ with $V(\Delta_a) = 2 + \tau^2(c^2\Delta_a - m^2)$ converges to $\frac{1}{c^2}\partial_t^2\phi - \nabla^2\phi + m^2\phi = 0$, with truncation error $\mathcal{O}(\tau^2, a^2)$ in the principal symbol (under the stated regularity).*

10.3 Dynamic envelope and effective metric

Let

$$s(x,t) = s_0 + \alpha\, S_\delta(0,x,t), \qquad \sigma = \frac{s}{\tau^2}, \qquad c_{\mathrm{eff}}^2 = c^2 - 2\sigma. \tag{9}$$

Then the wave equation becomes

$$\frac{1}{c^2}\partial_t^2\phi - \frac{c_{\mathrm{eff}}^2(x,t)}{c^2}\nabla^2\phi + m^2\phi = 0, \tag{10}$$

which is the KG equation in an effective metric $g^{00} = 1$, $g^{ij} = -(c_{\text{eff}}^2/c^2)\delta^{ij}$ up to gradients of c_{eff} (WKB). Null rays follow g-geodesics; weak-field identification $\sigma \simeq \gamma_\sigma \Phi$ reproduces Shapiro-like delays and lensing.

11 Quasi-static gravity and dynamic closure

Define the operational potential

$$\Phi(x) := \gamma_G \, \tau \, S_\delta(0, x). \tag{11}$$

In quasi-static regimes,

$$(\nabla^2 - \lambda_s^{-2})\Phi = \tilde{\kappa} \, S_\delta(0), \qquad \lambda_s \propto \sqrt{s}. \tag{12}$$

A dynamic closure respecting the speed bound is

$$\partial_t^2 \Phi + 2\Gamma \, \partial_t \Phi - c_{\text{eff}}^2 \nabla^2 \Phi + c_{\text{eff}}^2 \lambda_s^{-2} \Phi = \kappa_d \, \partial_t S_\delta(0, x, t). \tag{13}$$

12 APBC on a ring: a binary probe

On a ring of length L, PBC imposes $y_{n+L} = y_n$, APBC $y_{n+L} = -y_n$. APBC is equivalent to a Bloch-phase twist of $\Delta\theta = \pi/L$ per site, i.e., $\theta \mapsto \theta + \pi/L$. For the stride-$L$ decimated law, this induces $V_L \propto \cos(L\theta + \pi) = -\cos(L\theta)$, hence a *sign flip of* V_L while Q_L is unchanged. This cleanly diagnoses the clock sector.

13 Estimation pipeline (end-to-end)

Algorithm

Inputs: stream x_t, stride ℓ, (optional) spatial graph, envelope parameterization.
Outputs: (ρ, θ) or (V, Q); D; Φ; diagnostics.

1. Decimate: $y_k^{(r)} = x_{r+k\ell}$.
2. MOSC-filter and fit AR(2): estimate (V, Q); map to (ρ, θ) using (2).
3. Demodulate by r_+ and compute wrapped phase increments δ_t.
4. Estimate $S_\delta(0)$ via Abel/radial regularization; set $D = \frac{1}{2} S_\delta(0)$.
5. If spatial: choose L; set s (or $s(x)$); infer λ_s and $v_{\max}$.
6. Construct $\Phi = \gamma_G \tau S_\delta(0)$; solve screened Poisson (or dynamic closure).
7. Optional APBC probe: verify $V \mapsto -V$, Q unchanged.
8. Report metrics: $\mathcal{C}_2, N_{\text{eff}}, D, T_{\text{coh}}$ and residual diagnostics.

14 Diagnostics, units, and failure modes

Units. D has 1/time; V, Q are dimensionless. If L has units 1/length, then $\mathsf{L}^\dagger\mathsf{L}$ has 1/length^2 and the envelope $E = \exp(-s\,\mathsf{L}^\dagger\mathsf{L})$ requires s to have units of length^2. Then $\lambda_s \propto \sqrt{s}$ has units of length, and $\sigma := s/\tau^2$ has units of velocity^2, making $c_{\text{eff}}^2 = c^2 - 2\sigma$ dimensionally consistent.

Diagnostics. Dominant pair validity (AR(2) vs. AR(p)); MOSC stability ($|H|$ invariant); stationarity of δ_t; envelope influence monotonic in s; APBC flip check.

Failure modes. Multiple comparable pairs (need AR(p)); heavy phase slips; near-unit-circle singularities (regularize radially); graph anisotropy (reweight or continuum Laplace–Beltrami).

15 Glossary of non-standard terms

Term	Meaning
Residue track	A decimated subsequence $y_k^{(r)} = x_{r+k\ell}$ for fixed residue r modulo stride ℓ.
Reduced transfer	A 2×2 map M that advances a two-dimensional sufficient state across decimated steps; its $\mathrm{tr}, \det$ generate the exact two-step law via Cayley–Hamilton.
Wheel (AR(2) dictionary)	$(\rho, \theta) \leftrightarrow (V, Q)$ with $V = 2\rho\cos\theta$, $Q = \rho^2$; stride-ℓ generalization $V_\ell = 2\rho^\ell \cos(\ell\theta)$, $Q_\ell = \rho^{2\ell}$.
MOSC	Mirror–Outer Spectral Content: $\mathcal{S}[H] = H/I$ with I inner; preserves magnitude while removing all-pass phase.
Ω-compatibility	Equality of Abel-regularized spectral magnitudes for H and $\mathcal{S}[H]$.
APBC	Anti-periodic boundary condition: $y_{n+L} = -y_n$; induces $\theta \mapsto \theta + \pi/L$ and flips V_L at stride L while Q_L is unchanged.
Topological decoherence	Increase in $S_\delta(0)$ and local AR order due to APBC flips (boundary/topological changes), not just additive noise.
Operational potential	$\Phi = \gamma_G \tau S_\delta(0)$; in quasi-static limit obeys a screened Poisson equation.
Dynamic envelope	$s(x,t) = s_0 + \alpha S_\delta(0, x, t)$; renormalizes local speed and defines an effective metric.

A Cayley–Hamilton proof of the decimated two-step law

Let $u_k \in \mathbb{C}^2$ evolve as $u_{k+1} = M^\ell u_k$. Since $(M^\ell)^2 - (\mathrm{tr} M^\ell)M^\ell + (\det M^\ell)\mathrm{Id} = 0$, applying both sides to u_k and reading out $y_k = \alpha^\top u_k$ gives (1).

B MOSC factorization details

For rational H, the inner factor is a finite Blaschke product; dividing by it leaves an outer representative with identical boundary magnitude. Equality of Abel-regularized magnitudes follows from radial L^2 convergence.

C Dephasing decay derivation

Write $\phi_t = \sum_{j=0}^{t-1} \delta_j + \phi_0$. Then $\mathcal{C}(t) = \mathbb{E}[e^{i\sum_{j=0}^{t-1}\delta_j}]$. The cumulant expansion and weak dependence give $\log \mathcal{C}(t) \approx -\frac{1}{2}\mathrm{Var}(\sum_{j=0}^{t-1}\delta_j)$. Abel-regularizing the triangular covariance sum yields the rate $D = \frac{1}{2}S_\delta(0)$.

D Speed and screening derivation

Diagonalize in $\mathsf{L}^\dagger\mathsf{L}$-eigenmodes. The envelope multiplies mode λ by $e^{-s\lambda^2}$; group velocities are suppressed by at most $1/(1+s\lambda^2)$, giving $v_{\max} \lesssim v_0/(1+s\Lambda_{\max}^2)$ and $\lambda_s \propto \sqrt{s}$.

E Finite-loop holonomy bound (sketch)

Compare envelope-updated ordered products with the path-ordered exponential of the Levi–Civita connection via a non-Abelian Stokes estimate; bound the mismatch by curvature variation, discretization, and envelope bias terms.

F Green function on a ring

For net-zero sources on a ring of L sites with AR(2) parameters (V, Q), the Green function is

$$G_n^{(L)} = \frac{1}{L}\sum_{k=0}^{L-1} \frac{e^{2\pi i k n/L}}{V - 2\sqrt{Q}\cos(2\pi k/L)}, \qquad x = G^{(L)} * s. \tag{14}$$

G Ray tracing and Shapiro delay in an S_δ profile

G.1 Rays in the effective metric

With refractive index $n(x) = c/c_{\mathrm{eff}}(x)$, Fermat's principle yields $\frac{d}{ds}(n\,\dot{\boldsymbol{x}}) = \nabla n(\boldsymbol{x})$. For a weak Gaussian profile $\Phi(\boldsymbol{x}) = \Phi_0 e^{-\|\boldsymbol{x}\|^2/2R^2}$ and $n \approx 1 + \varepsilon e^{-r^2/2R^2}$ with $\varepsilon = \gamma_\sigma \Phi_0/c^2$,

$$\alpha(b) \approx \varepsilon\,\frac{\sqrt{2\pi}\,b}{R}\,e^{-b^2/2R^2}, \qquad \Delta T(b) \approx \frac{\varepsilon\sqrt{2\pi}\,R}{c}\,e^{-b^2/2R^2}. \tag{15}$$

G.2 Numerical ray tracer (pseudocode)

Listing 1: RK4 ray tracing in 2D with Gaussian $n(x)$

```
import numpy as np
```

```
def n_of_x(x, y, eps=1e-3, R=1.0):
    r2 = x*x + y*y
    return 1.0 + eps*np.exp(-0.5*r2/(R*R))

def grad_n(x, y, eps=1e-3, R=1.0):
    r2 = x*x + y*y
    coeff = -(eps*np.exp(-0.5*r2/(R*R)))/(R*R)
    return coeff*np.array([x, y])  # grad of n-1

def rk4_step(x, v, h, eps=1e-3, R=1.0):
    def accel(x, v):
        g = grad_n(x[0], x[1], eps, R)
        n = n_of_x(x[0], x[1], eps, R)
        vhat = v/np.linalg.norm(v)
        g_perp = g - np.dot(g, vhat)*vhat
        return g_perp / n
    k1x, k1v = v,                accel(x, v)
    k2x, k2v = v + 0.5*h*k1v,    accel(x + 0.5*h*k1x, v + 0.5*h*k1v)
    k3x, k3v = v + 0.5*h*k2v,    accel(x + 0.5*h*k2x, v + 0.5*h*k2v)
    k4x, k4v = v + h*k3v,        accel(x + h*k3x,     v + h*k3v)
    x_next = x + (h/6.0)*(k1x + 2*k2x + 2*k3x + k4x)
    v_next = v + (h/6.0)*(k1v + 2*k2v + 2*k3v + k4v)
    v_next = v_next/np.linalg.norm(v_next)
    return x_next, v_next
```


Acknowledgments

This master document unifies multiple drafts into a single coherent presentation with expanded explanations of all non-standard terms.